AMAY KASHYAP DEKA

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SUMMARY

High school senior working at the intersection of **computational oncology** and **clinical AI**, building models that connect biological structure, longitudinal patient data, and physician-facing decision support.

EDUCATION

Science Graduating March 2027
Delhi Public School, Guwahati **98.4%**, **1550** (SAT)
Don Bosco School, Guwahati
Relevant coursework:

HONORS AND AWARDS

Hackathon: Harvard Health Systems Innovation Lab Hackathon, Malaysia Hub Winner (1 of 2 winning teams)
Olympiads: Indian National Chemistry Olympiad (INChO) Qualifier and National Standard Examination in Chemistry (NSEC) State Topper (top **0.3%**); Indian National Mathematical Olympiad (INMO) Qualifier (top **1%**)
Scholarship: 100% Merit Scholarship, Delhi Public School (DPS) Guwahati (2 academic years)
Music: Associated Board of the Royal Schools of Music (ABRSM) Grade 5 Music Examination Certificate

TECHNICAL SKILLS

Bioinformatics & Computational Biology: Pan-cancer genomic data analysis, Raman spectroscopy, Oxford Nanopore metagenomics, molecular fingerprinting (Morgan), chemoinformatics (RDKit)
Machine Learning: PyTorch, TensorFlow, LangChain (RAG), recursive language models, regression models (Lasso, Ridge)
Programming: Python, R, TypeScript, AWS
Data Visualisation: Seaborn, Plotly, Matplotlib

PROFESSIONAL EXPERIENCE

SwiftDx, Guwahati, India: Founder & Chief Technology Officer (CTO), AI Surgical Pathology Platform April 2026-present

- Built (repo) virtual hematoxylin and eosin (H&E) staining and tumour-margin AI (Medical Open Network for AI (MONAI) margin assessment, confidence scoring, boundary overlays, Fast Healthcare Interoperability Resources (FHIR) reporting); patent application in process
- Named a **top 20 finalist globally**; pitched to venture capitalists (VCs) at Boston Health Week
- Won the **Harvard Health Systems Innovation Lab Hackathon** (Malaysia Hub, 1 of 2 winning teams); secured **4** pilots across **3** countries and validation with **4** university hospitals
- Collaborating with the Commonwealth Medical Association (medical centers in **56** countries)

Duong Lab, Montefiore Medical Center & Albert Einstein College of Medicine, New York City, New York: Clinical AI Co-Developer & Medical AI Researcher November 2025-present

- Co-built **RAGPhysician** (repo), a private desktop app for citation-backed clinical comma-separated values (CSV) queries (Electron, React, TypeScript, LangChain retrieval-augmented generation pipeline), now being piloted in the oncology department of Einstein School of Medicine
- Co-developed **Graph-RLM**, matching retrieval-augmented generation (RAG)'s **100%** test accuracy with $>$**97%** fewer tokens; among the first recorded uses of Recursive Language Models (RLMs) in a medical context
- Built RLM frameworks (Python, R) handling gigabytes of patient data to detect implicit/explicit follow-ups, and a novel triaging agent that removes manual routing of clinical data to sub-specialties
- Co-authored a radiology-focused large language model (LLM) review (multimodal imaging, report generation, hallucination risks); final review underway for submission to *European Journal of Radiology*

Research and Science Initiative (RSI), India: Raman Spectroscopy & Metagenomics Researcher Summer 2026

- Selected as **1 of 32** research scholars nationwide; research scholar under Professor Utpal Tatu, Chair of Biochemistry, Indian Institute of Science (IISc)
- Collaborated with HORIBA Labs, IISc and university biochemistry/animal husbandry departments; used Raman spectroscopy and statistical analysis to detect milk adulteration
- Studied the North-East (NE) Indian vaginal microbiome using Oxford Nanopore sequencing

Pratishruti, Guwahati, India: Technical Lead & Human Papillomavirus (HPV) Researcher June 2023-present

- Led an Indian Council of Medical Research (ICMR)-backed HPV vaccination drive securing **10K** Cervavac doses with income-based subsidies; coordinated ~**100** volunteers
- Made **80+** home visits to breast-cancer patients, spreading community awareness
- First author of the vaccination-drive case study and accompanying surveys, published in *Annals of Oncology Research and*

MycoStories, London, UK: Fungal Biotech Market Intelligence & Data Intern May 2025-August 2025

- Built a Python tool tracking **200+** fungal-biotech firms and **$2M+** in investments; findings shaped the company's market report

ACADEMIC PROJECTS

Graph RLM October 2025-present

Built a Graph-Recursive Language Model (Graph-RLM) framework for longitudinal breast oncology reasoning by converting patient records into directed clinical graphs and benchmarking graph-guided recursive reasoning against RAG, standard RLM, and full-context baselines.

- Designed an evidence-linked benchmark with **30** clinical questions across diagnosis, imaging, and procedure categories on electronic medical records (EMR), each tied to explicit supporting graph nodes for traceable evaluation.
- Achieved **100%** accuracy at **30.73** avg tokens/query for Graph-RLM versus **100%** at **1132.07** for RAG, **93.33%** at **3340.93** for standard RLM, and **96.67%** at **137451.83** for full-context prompting.

SMARCE1 Pan-Cancer Genomics Study March 2026-October 2026

Published in the Oxford Journal of Student Scholarship (OJSS); computational oncology and bioinformatics study of SMARCE1 expression across pan-cancer patient cohorts.

- Analyzed **148K+** patients across **12** pan-cancer studies and mapped SMARCE1 expression in **33** cancers, finding widespread overexpression, strongest in thymoma.

LEADERSHIP AND ACTIVITIES

Delhi Public School (DPS) Guwahati Health Awareness & Community Outreach Club Grades 11-12

- Founded and lead (President) a **50+** member health club; led **6** physician workshops, **4** awareness campaigns and **3** medical camps reaching **600+** students; trained **25+** volunteers

School Prefect, Don Bosco School Grade 10

- One of **4** Student Representatives leading **600+** students; organized **15+** inter-school/house events

Model United Nations (MUN) and Debate February 2024-present

- Competed at **25** MUNs, including Asia World MUN, Bali; won Best Delegate **6** times; chaired **200** delegates across **2** conferences; debate runner-up among **100+** participants
- Advanced to the global top 5 finals of the Harvard MissionBrain Hackathon with a rural stroke-care solution using interactive imagery on cigarette packets

MUSIC AND PERFORMING ARTS

Music and Cultural Leadership August 2020-present

- Passed **ABRSM Grade 5 Piano**; advanced to Grade 7 repertoire after skipping Grade 6; peer-tutored **2** pianists
- Earned **Visharad Part 2 (1st Division)** in Hindustani classical vocals at Bhatkhande Sangeet Vidyapith
- As Bihu Team Lead, led **25+** performers on vocals, pepa and taal to **2** interschool wins